Does an Agent-Based Model Provide Incremental Predictive Information In Macroeconomic Forecasting?

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Abstract

This thesis examines whether a large-scale agent-based model (ABM) forecast contains predictive information that is not already embedded in standard macroeconomic forecasting benchmarks. Using the publicly available Austrian forecasts of Poledna et al. (2023) for quarterly GDP growth and inflation, I compare the ABM signal to DSGE and reduced-form benchmarks (AR and a newly estimated BVAR), as well as a naive no-change forecast, across horizons up to 12 quarters. Beyond accuracy comparisons, I implement optimality diagnostics and forecast-encompassing regressions (pairwise and joint), using HAC inference for multi-step horizons and Holm–Bonferroni adjustments for multiple testing. The evidence suggests that the ABM forecast for inflation is not redundant and contributes incremental predictive content relative to competing benchmarks. The evidence for GDP growth forecasts does not meet conventional levels of statistical significance. The results are limited by weak statistical power but suggest that the ABM forecasts contain some incremental information at least some of the time. This complements previous findings on accuracy and highlights the potential value of ABMs as a forecasting tool. However, in addition to finding only weak evidence, the generalisability of these results is limited by the use of a single ABM, a single country, and a forecast window that contains an unusual macro environment.

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1 Introduction

Out-of-sample forecast evaluation is a critical tool for assessing macroeconomic models and their theoretical foundations. Agent-based models (ABMs) represent a promising departure from representative-agent frameworks, modelling the macroeconomy as the emergent outcome of heterogeneous interacting agents (Farmer 2025). While ABMs have been shown to reproduce certain stylized facts, their forecasting performance remains underexplored. To the best of my knowledge, there have been only four published attempts to use ABMs to forecast the macroeconomy. The first, by Delli Gatti and Grazzini (2020), was a proof-of-concept forecasting system built on a conceptual ABM of an economy with capital and credit. It underperformed relative to a Vector Autoregression (VAR) benchmark. The second, by Poledna et al. (2023), was a large-scale ABM of the Austrian economy, with each firm and person represented as an individual agent. It performed comparably to VAR and Dynamic Stochastic General Equilibrium (DSGE) benchmarks. The authors published the raw data and code (Poledna et al. 2022) which form the basis of the analysis in this thesis. The third, Wiese et al. (2024) extended the Poledna et al. (2023) ABM by applying Bayesian estimation techniques to improve calibration and forecast accuracy, and extended the analysis to all OECD countries. They reported substantial improvements over the original ABM and benchmarks for GDP, inflation, consumption and investments. Finally, Hommes et al. (2025) also extend the Poledna et al. (2023) ABM, introducing additional agent heterogeneity and benchmarking against the Bank of Canada's DSGE model. They reported superior performance for the ABM for GDP growth and consumption[1].

This existing work has focused primarily on point forecast accuracy. A closely related but distinct question is whether ABM forecasts provide unique information beyond that contained in the benchmarks. Models that are less or no more accurate may still be useful in practice if they contain information not captured by other models.

[1] No raw forecasts or code have been made available for the latter two.

This is especially relevant for ABMs, which are conceptually very different from the alternatives.

To answer this question, I evaluate whether the ABM forecasts from Poledna et al. (2023) provide unique information using forecast encompassing tests. I compare the ABM forecast to DSGE and AR benchmarks (also from Poledna et al. 2023), a Bayesian Vector Autoregression (BVAR) benchmark that I estimate myself, and a naive no-change forecast. I proceed in three steps. First, I briefly assess forecast accuracy. Second, I assess standard optimality diagnostics: unbiasedness, no autocorrelation, and unforecastability. Third, I conduct the forecast encompassing tests to answer the primary question.

This thesis contributes to the nascent ABM forecasting literature by providing the first comprehensive analysis of the information content of ABM forecasts relative to standard benchmarks.

The remainder of the thesis is structured as follows. Section 2 outlines the conceptual framework distinguishing forecast accuracy, optimality, and information content. Section 3 describes the data and Section 4 the models used. Section 5 presents the empirical results. Section 6 concludes, discusses limitations and suggests avenues for future research.

2 Theory: Forecast Evaluation Framework

2.1 Accuracy

The usefulness of forecasts can be evaluated along three complementary dimensions: accuracy, optimality, and information content. Accuracy is the most important objective measure of forecast quality. When having to choose a single forecast to rely on, any rational agent should prefer the most accurate forecast available. Accuracy is maximised when the forecast error (deviation between forecasted and realised values)

is minimised. A loss function defines a method for aggregating these errors. The most common loss functions are linear (treating the errors in absolute terms) and quadratic (weighting larger errors more heavily). Conceptually, quadratic loss functions define forecast optimality with respect to the conditional mean while linear loss functions define optimality with respect to the conditional median (Patton and Timmermann 2007). The Root Mean Squared Error (RMSE) is the most widely used accuracy metric under quadratic loss. Accuracy measures are important descriptive statistics, but they cannot tell us why forecasts differ.

2.2 Optimality

A second dimension of forecast evaluation is optimality. An optimal forecast is one that is internally consistent with rational prediction under the assumed loss function. Conceptually, we could say that an optimal forecast represents the best that could have been done given the information available at the time of forecasting (Muth 1961). As already stated, under squared-error loss, the optimal forecast is the conditional mean of the target variable. This implies that the forecast should be unbiased. "Conditional" refers to the information set available at the forecast origin. Therefore, forecast errors should be unpredictable using any information available at the forecast origin (this is referred to as efficiency). In this context, the information set includes all economic time series, other forecasts, and past forecast errors. I test the unbiasedness property and two types of efficiency (autocorrelation, efficiency with respect to past errors, and unforecastability, efficiency with respect to other available data[2]).

An important caveat is that forecast optimality is derived conditional on a stable data-generating process and information structure. Violations of optimality may therefore indicate either irrational forecasts (e.g. model misspecification or omitted information) or structural breaks (i.e. changes in the underlying data-generating

[2] Binder and Sekkel (2023) refer to these as weak-form and semi-strong-form efficiency respectively.

process). This is particularly relevant for macroeconomic forecasting, where structural instability is common (Clements 2005).

2.3 Information Content

Forecast information content is the third dimension of forecast evaluation considered here. It concerns whether a forecast contains predictive information that is not already present in other forecasts. In this framework, multiple forecasts are alternative, error-prone estimates of the same future outcome. When models rely on similar information sets or closely related structures, their forecast errors tend to be correlated, making their forecasts highly redundant. By contrast, models grounded in distinct theoretical or empirical approaches may exploit different features of the data-generating process and thus produce less correlated errors, thereby contributing complementary information.

Forecast encompassing tests formalise this idea by assessing whether one forecast subsumes the predictive information contained in another (Diebold 2006; Timmermann 2006). A standard joint encompassing test regresses the realised outcome on multiple competing forecasts (see Equation 11 in Section 5.3). If the coefficient on a given forecast is zero conditional on the others, that forecast provides no incremental predictive content. An optimal forecast with respect to the full information set cannot be encompassed. In practice, however, most forecasts are either not optimal, or optimal only with respect to restricted information sets. Such forecasts may therefore be encompassed, indicating redundancy, or fail to be encompassed, in which case they contribute distinct predictive information.

In this sense, a biased or inefficient forecast may still contribute useful information if other forecasts are biased or inefficient in different ways. Since no single model typically captures the full data-generating process and therefore encompasses all others, un-encompassed forecasts can be combined to exploit the distinct information

each provides. This approach is common in practice, where forecast combinations often outperform individual forecasts (Diebold 2006). From this perspective, the relevant question is not whether a forecast is optimal or most accurate in isolation, but whether it contributes incremental information that improves predictive performance when used alongside other forecasts.

More formally, let the full information set at forecast origin $t-h$ be denoted by $\mathcal{I}^F_{t-h}$. Each individual forecast i exploits a subset of this information, $\mathcal{I}^i_{t-h} \subseteq \mathcal{I}^F_{t-h}$. A forecast is therefore a mapping from its information set to a prediction of the future outcome:

$$\hat{x}^i_{t|t-h} = f^i(\mathcal{I}^i_{t-h}). \tag{1}$$

In principle, an idealised super-model would directly aggregate all available information sets,

$$\mathcal{I}^*_{t-h} = \bigcup_{i=1}^{N} \mathcal{I}^i_{t-h}, \tag{2}$$

and apply a single forecasting function to obtain the optimal forecast. Relative to this benchmark, generating separate forecasts and subsequently combining them entails a two-stage loss of information (Timmermann 2006). There are two practical obstacles to constructing such a super-model. First, pooling information sets may be infeasible (for example, when forecasters rely on private or proprietary information) or prohibitively costly (Diebold 2006). Second, expanding the information set increases model complexity and exacerbates parameter estimation problems (Timmermann 2006).

An alternative is therefore to combine forecasts rather than information sets. Let the information content of forecast i be represented by the signal it produces,

$$\mathcal{S}^i_{t-h} = \hat{x}^i_{t|t-h}. \tag{3}$$

A combined forecast (C) is then based on the collection of available forecast signals,

$$\mathcal{I}_{t-h}^{C} = \bigcup_{i=1}^{N} \mathcal{S}_{t-h}^{i}. \tag{4}$$

Not all forecast signals are necessarily informative once others are taken into account. A signal $\mathcal{S}_{t-h}^{i}$ is redundant relative to a set $\{\mathcal{S}_{t-h}^{j}\}_{j\neq i}$ if it provides no incremental predictive content for x_t beyond that contained in the remaining forecasts. Operationally, this corresponds to a forecast encompassing condition: forecast i is redundant if forecast errors cannot be reduced by conditioning on $\hat{x}_{t|t-h}^{i}$ once the other forecasts are included.

These distinctions motivate the empirical strategy of this thesis. An overview of forecast accuracy is reported in Section 5.1 and more detailed measures are reported in Appendix A, with a test for significant differences in accuracy between forecasts in Appendix B. Optimality diagnostics are presented in Section 5.2 to assess whether the forecasts satisfy rationality criteria. Since none of the forecasts considered are optimal with respect to the full information set, the analysis proceeds to forecast encompassing tests in Section 5.3. These tests address the central research question of the thesis: whether the agent-based model provides predictive information beyond that contained in benchmark forecasts.

3 Data

I evaluate pseudo out-of-sample quarterly forecasts for two core macroeconomic variables for Austria: real GDP growth and inflation, measured as growth in the GDP deflator. The forecasts and most underlying macro series are taken from the replication package of Poledna et al. (2023, 2022), which in turn draws primarily on Eurostat. The Eurostat extracts in the replication package were downloaded via the bulk download facility on or before May 2021 (based on package metadata) and therefore differ slightly

from the currently available data due to subsequent data revisions. The focus on GDP growth and inflation is motivated by their centrality to macroeconomic analysis. Other forecasting targets available in Poledna et al. (2023) are not considered here for brevity.

The underlying aggregate national accounts series come from Eurostat's `namq_10_gdp` dataset. Real GDP is chain-linked volumes with 2010 as the reference year measured in millions of euros. Both real GDP and the GDP deflator are seasonally and calendar adjusted. For forecast evaluation, I work with log differences (first differences of log values) of real GDP and of the GDP deflator, which approximate quarter-on-quarter percentage changes and support regression-based forecast evaluation in stationary form (Granger and Newbold 1974)[3]. Let Y_t denote the level of real GDP and P_t the level of the GDP deflator at time t. The realised (and forecasted) evaluation targets are:

$$y_t = \Delta \ln(Y_t) = \ln(Y_t) - \ln(Y_{t-1}) \approx \frac{Y_t - Y_{t-1}}{Y_{t-1}} \tag{5}$$

$$\pi_t = \Delta \ln(P_t) = \ln(P_t) - \ln(P_{t-1}) \approx \frac{P_t - P_{t-1}}{P_{t-1}} \tag{6}$$

For the ABM, DSGE, and AR models, I use forecasts directly from the Poledna et al. (2023) replication package. These forecasts are provided in level form and comprise a rectangular set of 28 forecast origins (2010Q1 through 2016Q4), each with 12-quarter-ahead forecasts, plus a triangular set of 11 origins (2017Q1 through 2019Q3) with progressively shorter forecast horizons due to a 2019Q4 terminal cut-off. In line with Poledna et al. (2023), I analyse horizons 1, 2, 4, 8, and 12 quarters ahead. At these horizons, the available number of forecast-actual pairs are 39, 38, 36, 32, and 28 respectively.

[3]ADF tests confirm that the actual values are stationary in this form. See Appendix C

The BVAR specification uses additional series from Poledna et al. (2023): household consumption, employee compensation, capital formation, and the household consumption deflator, also from Eurostat's `namq_10_gdp` dataset. The BVAR also incorporates five additional market-based indicators: the ATX (the main Austrian stock market index), a 3-month Euro Area interbank rate, an Austrian 10-year government bond yield, Brent crude oil prices, and the USD/EUR exchange rate. As they are publicly observable real-time indicators, they are not subject to revision. Oil prices are converted from USD to EUR using the USD/EUR exchange rate and deflated by the GDP deflator to obtain real prices. The USD/EUR series is backfilled to 1996 using the USD/DEM rate together with the irrevocable euro conversion rate (EUR/DEM$= 1.95583$). Daily and monthly series are converted to quarterly frequency using end-of-quarter values. Table 1 lists all sources and series codes. These eleven additional indicators are also converted to stationary form using log differences (except for interest rates and bond yields, which are converted to first differences).

Table 1: Data Sources and Series Codes

Indicator	Source	Dataset/Series Code
Primary variables (all models)		
Real GDP	Eurostat	`namq_10_gdp (B1GQ)`
GDP deflator	Eurostat	`namq_10_gdp (B1GQ)`
Additional national accounts variables (BVAR only)		
Real household consumption	Eurostat	`namq_10_gdp (P31_S14_S15)`
Household consumption deflator	Eurostat	`namq_10_gdp (P31_S14_S15)`
Real fixed capital formation	Eurostat	`namq_10_gdp (P51G)`
Employee compensation	Eurostat	`namq_10_gdp (D1)`
Additional market-based variables (BVAR and unforecastability tests)		
ATX stock index	Yahoo Finance	`^ATX`
3-month Euro Area interbank rate	OECD via FRED	`IR3TIB01EZM156N`
Austrian 10-year government bond yield	OECD via FRED	`IRLTLT01ATM156N`
Brent crude oil (USD)	EIA via FRED	`DCOILBRENTEU`
USD/EUR exchange rate (post-1999)	Fed via FRED	`DEXUSEU`
USD/DEM exchange rate (1996–1998)	FRBSL via FRED	`EXGEUS`

Notes: Eurostat series from `namq_10_gdp` are seasonally and calendar adjusted (SCA), measured in millions of euros (MEUR), for Austria (AT). Real values use chain-linked volumes with 2010 as the reference year (CLV10), nominal values use current prices (CP) and deflators use the price index (PD10_EUR). The series codes are given in brackets.

Table 2 provides summary statistics for actual GDP growth and inflation during

both the forecast period (2010–2019) and the training period (1996–2009).

Table 2: Summary Statistics for Actual GDP Growth and Inflation

Indicator	n	Mean	Std	Min	P5	P25	P50	P75	P95	Max
Panel A: Forecast Period (2010–2019)										
GDP	40	0.0038	0.0048	-0.0079	-0.0050	0.0011	0.0036	0.0067	0.0098	0.0164
Inflation	40	0.0042	0.0029	-0.0034	-0.0008	0.0035	0.0044	0.0055	0.0087	0.0112
Panel B: Training Period (1996–2009)										
GDP	55	0.0050	0.0065	-0.0190	-0.0066	0.0026	0.0054	0.0099	0.0129	0.0137
Inflation	55	0.0032	0.0048	-0.0079	-0.0055	0.0007	0.0039	0.0055	0.0102	0.0175

Notes: Summary statistics for actual quarterly GDP growth and inflation during the forecast evaluation period (2010Q1–2019Q4, Panel A) and model training period (1996Q1–2009Q4, Panel B). Values are log differences, which approximate quarterly growth rates. For example, a value of 0.005 represents approximately 0.5 percentage points of quarterly growth. Note that log differencing loses the first observation (1996Q1), resulting in n=55 observations for the training period. P5, P25, P50, P75, P95 denote percentiles.

Comparing the two panels reveals that the forecast period (2010–2019) experienced somewhat lower volatility for both indicators compared to the training period (1996–2009). The training period includes the global financial crisis, while the forecast period captures the European debt crisis and the gradual recovery and stabilisation that followed. Both the mean and the median GDP growth was higher over the whole training period, while mean inflation was lower.

4 Models

I extend the forecast evaluation in Poledna et al. (2023) by examining quarterly forecasts from 2010Q2 to 2019Q4 for five forecasting models representing different methodological approaches. Three models come from the original paper: an agent-based model (ABM), a dynamic stochastic general equilibrium model (DSGE), and an autoregressive benchmark (AR). I add two additional models: a Bayesian vector autoregression (BVAR) and a naive no-change benchmark. Together, these models span the spectrum from microfounded structural models to reduced-form time series methods and a simple persistence benchmark. The choice of such a diverse

set is motivated by the finding that no single approach consistently dominates in macroeconomic forecasting. In different sample periods and for different indicators BVAR, AR and DSGE models have all been found to be the most accurate (Gürkaynak et al. 2013).

The agent-based model from Poledna et al. (2023) is specifically designed for macroeconomic forecasting and calibrated to the Austrian economy using microdata. The model incorporates heterogeneous agents including households, firms, banks, and government, with decentralized markets. The simulation develops across 14 phases for each timestep (see Table 3). Each forecast is based on the mean outcome of 500 Monte Carlo simulations.

Table 3: ABM Simulation Phases

Phase	Description
1	Expectation formation
2	Exogenous shocks
3	Monetary policy
4	Firm planning
5	Credit market
6	Labour market
7	Production
8	Household planning
9	Government planning
10	Trade supply and demand
11	Goods market
12	Inflation
13	Accounting
14	Bankruptcy resolution

Notes: This table shows the phases the ABM simulation goes through in each timestep, based on the code in Poledna et al. (2022).

The DSGE model from Poledna et al. (2023) is a medium-scale small open economy model (adapted from Smets and Wouters 2007) built on microfoundations with representative agents optimizing intertemporally under rational expectations. It includes nominal rigidities, habit formation, investment adjustment costs, and a Taylor-rule monetary policy.

The AR specification from Poledna et al. (2023) was estimated separately for each

indicator using Maximum Likelihood with lag length selected by information criteria (lag length 1 was selected in all cases).

I introduce a Bayesian vector autoregression (BVAR) as an additional multivariate benchmark. BVARs extend AR models by jointly modelling multiple series and using shrinkage priors to mitigate overfitting while capturing cross-variable dynamics. I estimate a lag order 1 stationary BVAR on 11 variables with three-level hierarchical priors following Giannone et al. (2015).

The 11 variables include six series from Poledna et al. (2023): real GDP, household consumption, employee compensation, real fixed capital formation, and the GDP and household consumption deflators. I add five financial indicators: the ATX stock index and Austrian 10-year bond yields, plus 3-month Euro Area interbank rates, Brent oil prices (real EUR), and the EUR/USD exchange rate. Bond yields and interest rates are in first differences, all other series are in log differences.

The choice of variables is motivated by two considerations. First, the BVAR should include key national accounts aggregates as well as forward-looking market-based indicators informative about GDP and inflation. Second, given the limited sample size, the number of variables is kept relatively small, and the series are transformed to improve stationarity and estimation stability.

As a basic diagnostic for redundant information, I examine pairwise correlations in the transformed data and would exclude variables exhibiting near-duplicate co-movement (absolute correlation above 0.9). In the transformed series, all pairwise correlations are below 0.6, so no candidates are removed.

Estimation uses an expanding window starting in 1996 and re-estimates the BVAR at each forecast origin. For each estimation, I run a 5,000-draw burn-in followed by 50,000 posterior draws.

I produce forecasts using the same origins and horizons as the other models. Forecasts are iterated posterior predictive simulations, and the reported point forecast

is the posterior mean.

Finally, the naive benchmark assumes no change from the previous period's value, providing a minimal baseline to assess whether structural or multivariate models add predictive value beyond simple persistence. For the log-differenced series used in this thesis, the naive forecast corresponds to the assumption that the current quarter's GDP growth rate and inflation will persist into all future quarters.

5 Results: Does the ABM Forecast Add Information?

5.1 Accuracy Overview

Before turning to optimality and information-content diagnostics, I establish whether the competing forecasts are at least competitive in conventional point-accuracy terms. Table 4 summarises forecast accuracy using Theil's U_2 statistic (forecast RMSE divided by naive RMSE, see equation A.2) averaged across horizons, while Appendix A reports horizon-specific U_2 values and raw RMSE levels. All forecasts outperform the naive no-change benchmark for both GDP growth and inflation at all evaluated horizons ($U_2 < 1$ throughout), with averages well below 1.0. The BVAR shows the strongest average performance for GDP (0.623), while for inflation differences are minimal across ABM, BVAR, and AR (all around 0.66), with DSGE somewhat weaker on this metric in the present sample.

Table 4: Theil's U_2 Statistic: Average Across Horizons

Indicator	ABM	DSGE	BVAR	AR	Naive
GDP	0.651	0.654	0.623	0.671	1.000
Inflation	0.661	0.725	0.665	0.665	1.000

Notes: Average of Theil's U_2 (forecast RMSE divided by naive RMSE) across forecast horizons $h \in \{1, 2, 4, 8, 12\}$. Values below 1 indicate the forecast outperforms a naive no-change benchmark. See Table A.2 for horizon-specific values.

However, despite this consistent improvement over naive, these differences do not rise to the level of statistical significance in the Modified Diebold-Mariano (MDM) comparisons after multiple-testing adjustments[4] (unadjusted MDM test results are reported in Appendix B). Overall, point-accuracy differences are modest and both indicator- and horizon-dependent.

5.2 Optimality Diagnostics

I test whether the forecasts satisfy three optimality conditions implied by rationality under squared-error loss. Such conditions are often violated in practice (Binder and Sekkel 2023). Evidence against these conditions would suggest that forecast combination may yield gains.

First, forecast unbiasedness is assessed using the Mincer-Zarnowitz regression (Mincer and Zarnowitz 1969):

$$x_t = \beta_0 + \beta_1 \hat{x}_{t|t-h} + u_{t|t-h}, \qquad H_0 : \ \beta_0 = 0, \ \beta_1 = 1 \tag{7}$$

Table 5 reports mean forecast bias alongside Holm-Bonferroni adjusted $\tilde{p}$-values from the Mincer-Zarnowitz tests. All models show statistically significant bias on at least 3 (of 10) horizon-indicator combinations after multiple testing correction, indicating systematic forecast errors that persist even under conservative inference[5]. The naive forecast fails the unbiasedness test in every case (despite a low mean bias) because the forecasts do not co-move appropriately with realized values (β_1 deviates from 1). Because the multiple-testing corrections reduce the power of these tests substantially, a failure to reject the null should not be interpreted as positive evidence

[4]Appendix D details the purpose and method of applying Holm-Bonferroni multiple-testing adjustments.

[5]For additional discussion of the methodology of these tests, see Appendix E.

for unbiasedness.

Table 5: Mean Forecast Bias

Model	Horizon	GDP	Inflation
ABM	$h = 1$	-0.0023 (0.0135**)	-0.0004 (0.2788)
	$h = 2$	-0.0016 (0.6194)	-0.0008 (0.0735*)
	$h = 4$	-0.0012 (0.1444)	-0.0007 (1.0000)
	$h = 8$	-0.0011 (0.5716)	-0.0008 (1.0000)
	$h = 12$	-0.0016 (0.0890*)	-0.0004 (1.0000)
DSGE	$h = 1$	-0.0026 (0.1411)	0.0003 (0.0000***)
	$h = 2$	-0.0023 (0.2196)	-0.0002 (0.0108**)
	$h = 4$	-0.0020 (0.6182)	-0.0008 (0.0340**)
	$h = 8$	-0.0016 (0.3110)	-0.0007 (0.6286)
	$h = 12$	-0.0021 (0.6286)	0.0005 (0.0151**)
BVAR	$h = 1$	0.0002 (0.6286)	-0.0001 (1.0000)
	$h = 2$	0.0006 (1.0000)	-0.0005 (0.0662*)
	$h = 4$	0.0010 (0.1980)	-0.0007 (0.1365)
	$h = 8$	0.0012 (0.0000***)	-0.0007 (0.0159**)
	$h = 12$	0.0008 (0.0000***)	-0.0004 (1.0000)
AR	$h = 1$	-0.0022 (0.3767)	-0.0005 (0.0505*)
	$h = 2$	-0.0018 (0.2879)	-0.0008 (1.0000)
	$h = 4$	-0.0017 (0.3533)	-0.0007 (0.6286)
	$h = 8$	-0.0016 (0.0380**)	-0.0007 (0.1745)
	$h = 12$	-0.0022 (0.0154**)	-0.0003 (1.0000)
Naive	$h = 1$	-0.0003 (0.0000***)	-0.0001 (0.0000***)
	$h = 2$	0.0001 (0.0000***)	-0.0003 (0.0000***)
	$h = 4$	0.0004 (0.0000***)	-0.0002 (0.0000***)
	$h = 8$	0.0004 (0.0008***)	-0.0002 (0.0000***)
	$h = 12$	-0.0004 (0.0000***)	0.0003 (0.0000***)

Notes: This table shows mean forecast bias ($\hat{x}_{t|t-h} - x_t$). In parentheses: Holm-Bonferroni adjusted $\tilde{p}$-values from the Mincer-Zarnowitz joint F test for unbiasedness with HAC standard errors (see Equation 7). Significance (rejecting unbiasedness assumption): *** $\tilde{p} < 0.01$, ** $\tilde{p} < 0.05$, * $\tilde{p} < 0.10$.

The economic significance of the biases reported in Table 5 differs considerably between GDP and inflation. Since both indicators are measured as log differences, the bias values can be interpreted approximately as percentage points (pp). A log difference of 0.005 corresponds to roughly 0.5pp of quarterly growth.

For GDP, the biases are economically substantial. As shown in Table 2, quarterly GDP growth for Austria averaged around 0.4pp during the sample period (ranging from -0.8pp to +1.6pp). A systematic bias of -0.2pp therefore represents roughly half the average growth rate, a significant underestimation. The Naive benchmark exhibits

less mean bias than the structural models (ABM, DSGE) for GDP forecasting, with only BVAR approaching Naive performance at short horizons.

For inflation, the mean biases are more modest. With average quarterly inflation also around 0.4pp (Table 2), observed biases on the order of -0.04pp (for example for the ABM at h=1) represent roughly one-tenth of the average rate, meaningful but less severe than the GDP forecast biases.

Second, weak forecast efficiency tests examine whether errors contain predictable autocorrelation beyond the mechanical dependence induced by overlapping horizons. For h-step-ahead forecasts, autocorrelation up to lag $h-1$ arises mechanically, so I focus on autocorrelation at lag h.

$$\rho_h = \frac{\mathrm{Cov}(e_{t|t-h}, e_{t-h|t-2h})}{\sigma_{e_{t|t-h}} \sigma_{e_{t-h|t-2h}}}, \qquad H_0 : \ \rho_h = 0 \tag{8}$$

Table 6 provides no evidence of statistically significant autocorrelation at lag h for any model or horizon (except the naive inflation forecasts), consistent with the assumption that multi-step forecast errors behave approximately as an MA$(h-1)$ process. Given the short evaluation window, however, these are low-power diagnostics and should be interpreted cautiously.

Third, unforecastability tests examine whether forecast errors can be predicted by other information available at the forecast origin. I examine whether forecast errors can be predicted by several market-based variables: oil prices (in Real EUR), EUR/USD exchange rate, ATX stock market index, 3-month Euro Area interbank interest rate, Austrian 10-year government bond yield (see Section 3 for more details).

For each model i, I regress forecast errors on these variables:

$$e^i_{t|t-h} = \alpha + \sum_{k=1}^{K} \beta_k z_{k,t-h} + u_{t|t-h}, \qquad H_0 : \ \beta_1 = \beta_2 = \cdots = \beta_K = 0. \tag{9}$$

Table 6: Weak Efficiency Tests: Autocorrelation

Model	Horizon	GDP	Inflation
ABM	$h = 1$	0.201 (1.0000)	-0.242 (1.0000)
	$h = 2$	0.207 (1.0000)	-0.077 (1.0000)
	$h = 4$	0.118 (1.0000)	-0.334 (1.0000)
	$h = 8$	0.158 (1.0000)	-0.102 (1.0000)
	$h = 12$	-0.362 (1.0000)	-0.115 (1.0000)
DSGE	$h = 1$	-0.060 (1.0000)	-0.370 (0.9833)
	$h = 2$	0.107 (1.0000)	-0.001 (1.0000)
	$h = 4$	0.153 (1.0000)	-0.295 (1.0000)
	$h = 8$	0.206 (1.0000)	-0.147 (1.0000)
	$h = 12$	-0.336 (1.0000)	-0.124 (1.0000)
BVAR	$h = 1$	-0.027 (1.0000)	-0.072 (1.0000)
	$h = 2$	0.094 (1.0000)	-0.107 (1.0000)
	$h = 4$	0.119 (1.0000)	-0.305 (1.0000)
	$h = 8$	0.148 (1.0000)	-0.098 (1.0000)
	$h = 12$	-0.358 (1.0000)	-0.125 (1.0000)
AR	$h = 1$	0.194 (1.0000)	-0.257 (1.0000)
	$h = 2$	0.226 (1.0000)	-0.073 (1.0000)
	$h = 4$	0.111 (1.0000)	-0.334 (1.0000)
	$h = 8$	0.160 (1.0000)	-0.084 (1.0000)
	$h = 12$	-0.329 (1.0000)	-0.129 (1.0000)
Naive	$h = 1$	-0.427 (0.3696)	-0.510 (0.0725*)
	$h = 2$	-0.202 (1.0000)	-0.344 (1.0000)
	$h = 4$	-0.234 (1.0000)	-0.525 (0.0914*)
	$h = 8$	-0.255 (1.0000)	-0.405 (1.0000)
	$h = 12$	-0.213 (1.0000)	-0.339 (1.0000)

Notes: Forecast efficiency measured by autocorrelation at the critical lag h (ρ_h). Holm-Bonferroni adjusted $\tilde{p}$-values in parentheses. Significance: * $\tilde{p} < 0.10$.

Here, $z_{k,t-h}$ denotes variable k observed at time $t-h$. Rejection of H_0 in an F-test implies that forecast errors are predictable from information available at the forecast origin. The results of these tests are shown in Table 7.

Table 7: Unforecastability Tests

Model	Horizon	GDP	Inflation
ABM	$h=1$	7.72 (0.0034***)	0.72 (1.0000)
	$h=2$	3.68 (0.2892)	1.60 (1.0000)
	$h=4$	7.21 (0.0071***)	3.17 (0.4736)
	$h=8$	8.23 (0.0045***)	2.80 (0.7109)
	$h=12$	4.86 (0.1413)	3.76 (0.3665)
DSGE	$h=1$	3.68 (0.2889)	1.69 (1.0000)
	$h=2$	2.58 (0.8141)	2.01 (1.0000)
	$h=4$	6.50 (0.0146**)	4.12 (0.1962)
	$h=8$	6.13 (0.0289**)	3.16 (0.5092)
	$h=12$	2.33 (1.0000)	4.62 (0.1783)
BVAR	$h=1$	2.32 (1.0000)	0.20 (1.0000)
	$h=2$	4.53 (0.1182)	0.73 (1.0000)
	$h=4$	6.32 (0.0175**)	3.36 (0.3941)
	$h=8$	7.41 (0.0088***)	2.12 (1.0000)
	$h=12$	3.56 (0.3943)	3.68 (0.3811)
AR	$h=1$	7.11 (0.0062***)	0.81 (1.0000)
	$h=2$	3.55 (0.3328)	1.43 (1.0000)
	$h=4$	6.16 (0.0205**)	2.79 (0.7000)
	$h=8$	8.03 (0.0052***)	2.41 (1.0000)
	$h=12$	3.19 (0.5428)	4.13 (0.2734)
Naive	$h=1$	0.61 (1.0000)	0.93 (1.0000)
	$h=2$	0.39 (1.0000)	1.32 (1.0000)
	$h=4$	3.44 (0.3811)	0.42 (1.0000)
	$h=8$	4.31 (0.1910)	4.91 (0.1057)
	$h=12$	5.63 (0.0692*)	4.48 (0.1962)

Notes: Tests whether forecast errors can be predicted by several market-based variables available at the forecast origin. Values shown are F-statistics (testing the null that all coefficients are zero) with HAC standard errors. Holm-Bonferroni adjusted $\tilde{p}$-values in parentheses. Significance: *** $\tilde{p} < 0.01$, ** $\tilde{p} < 0.05$, * $\tilde{p} < 0.10$.

Except for the naive model, all models have forecastable errors in their GDP forecasts for 2-3 out of 5 horizons. For inflation forecasts the evidence for forecastable error does not pass conventional significance levels. Because of the conservative multiple-testing adjustments, however, failure to reject should not be interpreted as positive evidence of unforecastability.

Unforecastability rejections indicate that models do not fully incorporate readily available market information at the forecast origin. This does not by itself imply that forecasts are complementary, but it motivates encompassing tests: if different models omit different components of the relevant information set, their signals need not be redundant once observed jointly.

5.3 Forecast Encompassing

To directly assess whether the ABM contributes incremental predictive information, I employ forecast encompassing regressions following Diebold (2006). In the terminology of Section 2, these tests examine whether the forecast signal generated by one model is redundant conditional on the signal generated by another.

Forecast-encompassing regressions are informative about redundancy versus incremental content, but inference is challenging when competing forecasts are highly correlated (which is expected for good forecasts). In such cases, coefficient estimates can be imprecise and hypothesis tests have low power, so non-rejection should not be read as evidence of equivalence.

For each pair of models (a, b), I estimate:

$$x_t = \beta_a \hat{x}^a_{t|t-h} + \beta_b \hat{x}^b_{t|t-h} + u_{t|t-h}, \qquad H_0 : \ \beta_a = 0 \tag{10}$$

and test whether each forecast contains incremental predictive information conditional on the other. In particular, if $\beta_a = 0$, the signal $\hat{x}^a_{t|t-h}$ is redundant given $\hat{x}^b_{t|t-h}$ and does not contribute additional predictive content for x_t. Excluding an intercept restricts admissible forecast combinations to the linear span of the individual forecasts and corresponds to the approach of Diebold (2006).[6]

Table 8 reports the estimated coefficients β_a from pairwise encompassing regressions

[6]The with-intercept specification of Timmermann (2006), which allows for ex post bias correction, is examined as a robustness check in Appendix F.

in which the ABM is treated as model a against each benchmark model. In this specification, the ABM provides statistically significant incremental information in 7 out of 20 comparisons for inflation (Panel B), but only 2 out of 20 for GDP (Panel A).

Table 8: Pairwise Encompassing Tests (No Intercept): ABM vs Other Models

Other Model	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 12$
Panel A: GDP					
DSGE	1.69 (0.224)	1.18 (0.374)	0.47 (1.000)	-1.20 (1.000)	-1.38 (0.312)
BVAR	1.30 (1.000)	0.51 (1.000)	-2.08 (1.000)	-1.74 (1.000)	-2.75 (1.000)
AR	-2.67 (1.000)	3.19 (0.878)	0.75 (1.000)	1.74 (1.000)	2.54 (1.000)
Naive	2.07 (0.207)	1.52 (0.067*)	1.46 (0.446)	1.63 (0.060*)	1.52 (0.953)
Panel B: Inflation					
DSGE	1.28 (0.224)	1.16 (0.037**)	1.24 (0.026**)	0.60 (0.707)	1.60 (0.235)
BVAR	0.41 (1.000)	1.35 (1.000)	2.25 (0.707)	4.61 (0.646)	2.71 (1.000)
AR	0.55 (1.000)	-1.97 (1.000)	3.40 (0.290)	2.94 (1.000)	4.25 (0.267)
Naive	1.36 (0.002***)	1.32 (0.000***)	1.53 (0.000***)	1.34 (0.000***)	0.98 (0.000***)

Notes: Pairwise forecast encompassing test, see equation 10. The table reports the estimated coefficient on the ABM forecast from a regression including the ABM and the indicated benchmark model. The null hypothesis is $\beta_a = 0$, implying that the ABM provides no incremental predictive information conditional on the benchmark forecast. Holm–Bonferroni adjusted $\tilde{p}$-values in parentheses. Significance: *** $\tilde{p} < 0.01$, ** $\tilde{p} < 0.05$, * $\tilde{p} < 0.10$.

I also conduct a joint test of incremental predictive content by including all K forecasts simultaneously:

$$x_t = \sum_{i=1}^{K} \beta_i \hat{x}^i_{t|t-h} + u_{t|t-h}, \qquad H_0 : \ \beta_i = 0 \tag{11}$$

This specification imposes a more demanding encompassing criterion. Forecasts that are not redundant in pairwise comparisons may nevertheless become redundant once the full set of competing forecast signals is considered.

The results of the joint encompassing regressions are reported in Table 9, with the first row presenting results for GDP and the second row for inflation. After applying the Holm–Bonferroni correction for multiple testing, the ABM contributes statistically significant incremental information in 1 out of 5 horizons for inflation (h=8, $\tilde{p} = 0.023$) and never for GDP.

The joint encompassing analysis provides additional evidence that the ABM

Table 9: Joint Encompassing Tests (No Intercept): ABM Coefficients

Indicator	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 12$
GDP	-1.50 (1.000)	2.06 (1.000)	-1.10 (1.000)	-1.10 (1.000)	-0.92 (1.000)
Inflation	0.24 (1.000)	-1.64 (1.000)	3.70 (0.244)	4.54 (0.023**)	3.10 (1.000)

Notes: ABM coefficients from joint forecast encompassing test, see equation 11. The table reports estimated coefficient β_i, where model i is the ABM, across forecast horizons h for GDP and inflation. The null hypothesis is $\beta_i = 0$, implying that the ABM provides no incremental predictive information conditional on all other forecasts. Holm–Bonferroni adjusted $\tilde{p}$-values in parentheses. Significance: *** $\tilde{p} < 0.01$, ** $\tilde{p} < 0.05$, * $\tilde{p} < 0.10$.

contributes unique predictive information in forecasting inflation for at least one horizon and may therefore be profitably included in forecast combinations.

6 Conclusions and Outlook

Building on the agent-based macroeconomic forecasting framework of Poledna et al. (2023) and subsequent applications, this thesis asks whether the ABM forecast contains incremental predictive information beyond established benchmarks.

I find evidence that it does: for inflation, the ABM provides incremental predictive content in at least 35% of ABM–benchmark combinations (across horizons) in the no-intercept encompassing specification (conditional on the competing model forecasts), although inference is constrained by the short evaluation window. The evidence for GDP mostly does not reach statistical significance after multiple-testing adjustment.

The remainder of this conclusion summarises the pattern of results across accuracy, optimality diagnostics, and encompassing tests, and discusses implications, limitations, and avenues for future research.

In macroeconomic forecasting, BVAR, DSGE, and AR models have all been found to outperform each other in different contexts and samples (Gürkaynak et al. 2013). The present results are consistent with this finding: no single model dominates across all horizons and indicators in terms of point accuracy.

While accuracy differences are not statistically significant, the accuracy results suggest that all models are consistently better than the naive no-change benchmark but have comparable accuracy among themselves.

In terms of optimality, all models exhibit statistically significant bias in some combinations of horizon and indicator. Autocorrelation tests provide no evidence against weak efficiency (except for the naive forecast). All models have forecastable errors, violating semi-strong efficiency. These rejections imply that no model fully subsumes the available information set, motivating the encompassing analysis as the appropriate test for redundancy versus incremental information.

In the encompassing tests, the ABM provides statistically significant incremental predictive content in some cases for inflation forecasts. The strict family-wise error control via Holm-Bonferroni adjustment means that the reported significant results are robust to multiple testing concerns. Although the evidence is not overwhelming, the results suggest that the ABM signal is not redundant for inflation.

Appendix G reports unadjusted encompassing test results for all model pairs. Since these are not adjusted for multiple testing, no conclusions should be drawn from the p-values individually.

Macroeconomic forecasting is difficult, and simple Autoregressive models are hard to beat. Since such forecasts are unlikely to be optimal in practice, it may be worthwhile to include other, more sophisticated models in the forecast combinations to extract any incremental information they may provide. Traditionally, DSGE and BVAR models have been suitable candidates for such combinations. The present results tentatively suggest that ABM forecasts may also be useful additions to such combinations, especially for inflation forecasting. Such combinations may be particularly relevant in contexts where inflation dynamics are complex and driven by heterogeneous agent behaviour, which ABMs are well-suited to capture.

Outside of forecasting practice, the results also have implications for model

evaluation and ABM research. Encompassing tests provide a more nuanced view of model performance than point-accuracy rankings alone. Incremental information content in ABM forecasts supports the claim that ABMs can represent macroeconomically relevant microfoundations not captured by other models.

However, several limitations constrain the generalisability and practical relevance of these findings.

First, the short evaluation window (2010Q2–2019Q4) limits statistical power and may not capture performance across different macro regimes. This period is characterised by prolonged interest rate accommodation and subdued inflation accompanied by moderate growth. Extending the evaluation period or pooling across horizons (e.g., Clements et al. 2007) could increase power and robustness. For example, extending the sample to include more recent periods, including the pandemic and high-inflation episodes, would allow assessment of whether the identified incremental information content persists across different macro regimes. The existing extensions by Wiese et al. (2024) and Hommes et al. (2025) do not extend into this period, so new forecast series would need to be generated. Aside from increasing the generalisability of the results, this would also increase the sample size and therefore statistical power to detect incremental information content.

Second, the external validity is limited by the focus on a single country (Austria) and a single ABM specification. The generalisability of the findings to other economies or ABM architectures is therefore uncertain. Replicating the analysis across countries and ABM variants would help assess external validity. The work of Wiese et al. (2024) is particularly relevant here, as they develop a Bayesian-calibrated successor to the ABM used in this thesis for all OECD countries (with an added housing market, but fewer industries and a 1:1,000 scale reducing computational needs dramatically). When forecasts from that model become available, applying the present encompassing framework would be a natural next step. This is especially important given that

ABMs represent so much microstructure that will differ from country to country. For this reason, extending this analysis to Hommes et al. (2025) Canadian ABM, which has additional agent heterogeneity and some minor Canada-specific changes, would allow an additional comparison of incremental predictive content and further improve external validity.

Third, the reliance on historical microdata with long publication delays (3-4 years) limits the real-time feasibility of these ABM forecasts. This constrains their operational forecasting relevance and comparability to real-time benchmarks. Developing proxy or nowcast methods for the required micro inputs, quantifying proxy error propagation, and explicitly evaluating real-time feasibility are important next steps. Once real-time forecasting is achieved, some plausible benchmark candidates for an Austrian ABM could be forecasts by the Austrian National Bank (OeNB), the Institute for Advanced Studies (IHS) and the Austrian Institute of Economic Research (WIFO) (Fortin et al. 2020). Such a comparison would directly test whether the incremental inflation signal identified here persists once the conditioning set reflects the forecasts that policy makers actually observe.

Looking forward, the most valuable extensions are those that strengthen inference about incremental information content and move the analysis closer to operational forecasting settings. I have detailed several such extensions above.

Finally, a methodological extension would broaden what is meant by incremental information. Extending the framework to density forecasts would allow the ABM's empirically realistic fat-tailed distributions (Ascari et al. 2015) to be evaluated directly against recent DSGE specifications that replicate similar behaviour (Dave and Sorge 2025). Coroneo et al. (2024) recently demonstrated a forecast encompassing method for continuous-variable density forecast encompassing.

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Appendices

A Measures of Forecast Accuracy

I evaluate forecast accuracy using several complementary metrics. Root mean squared error (RMSE, Table A.1) measures the average magnitude of forecast errors in the original units of the series, providing a scale-dependent accuracy measure that penalizes large errors more heavily than small ones:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(x_t - \hat{x}_{t|t-h})^2} \tag{A.1}$$

Table A.1: Root Mean Squared Error (RMSE)

Model	Indicator	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 12$
ABM	GDP	0.0051	0.0044	0.0043	0.0041	0.0038
	Inflation	0.0030	0.0028	0.0028	0.0028	0.0025
DSGE	GDP	0.0051	0.0046	0.0044	0.0039	0.0038
	Inflation	0.0034	0.0031	0.0031	0.0028	0.0029
BVAR	GDP	0.0048	0.0041	0.0043	0.0041	0.0035
	Inflation	0.0029	0.0028	0.0029	0.0029	0.0025
AR	GDP	0.0050	0.0046	0.0045	0.0043	0.0041
	Inflation	0.0030	0.0027	0.0028	0.0029	0.0025
Naive	GDP	0.0066	0.0060	0.0070	0.0075	0.0067
	Inflation	0.0042	0.0041	0.0046	0.0044	0.0038

Notes: See Equation A.1. Lower values indicate better accuracy.

Theil's U_2 statistic (Table A.2) normalizes RMSE relative to a naive no-change benchmark, with values below one indicating that the model outperforms simple persistence:

$$U_2 = \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{naive}}} = \frac{\sqrt{\frac{1}{n}\sum_{t=1}^{n}(x_t - \hat{x}_{t|t-h})^2}}{\sqrt{\frac{1}{n}\sum_{t=1}^{n}(x_t - x_{t-h})^2}} \tag{A.2}$$

where $U_2 < 1$ indicates the model outperforms naive persistence, $U_2 = 1$ indicates equal performance, and $U_2 > 1$ indicates underperformance.

Table A.2: Theil's U_2 Statistic

Model	Indicator	$h=1$	$h=2$	$h=4$	$h=8$	$h=12$
ABM	GDP	0.772	0.739	0.625	0.548	0.573
	Inflation	0.724	0.679	0.603	0.640	0.658
DSGE	GDP	0.775	0.768	0.629	0.526	0.569
	Inflation	0.820	0.746	0.669	0.630	0.757
BVAR	GDP	0.734	0.695	0.612	0.551	0.525
	Inflation	0.699	0.689	0.626	0.649	0.661
AR	GDP	0.755	0.764	0.643	0.572	0.618
	Inflation	0.726	0.663	0.617	0.648	0.670
Naive	GDP	1.000	1.000	1.000	1.000	1.000
	Inflation	1.000	1.000	1.000	1.000	1.000

Notes: Theil's U_2 compares forecast accuracy to a naive no-change forecast: $U_2 < 1$ indicates the forecast is better than naive, $U_2 = 1$ equivalent, $U_2 > 1$ worse than naive. Lower values indicate better accuracy.

A curious result is that both the RMSE and U_2 appear to decrease with forecast horizon for the AR, ABM and DSGE models (for both indicators), which is counterintuitive as longer-horizon forecasts typically exhibit lower accuracy due to increased uncertainty. To investigate this, I compute the sample standard deviation of the target variable at each horizon and a scaled RMSE (sRMSE) that normalizes RMSE by that standard deviation:

$$\widehat{\sigma}_x(h) = \sqrt{\frac{1}{N_h - 1} \sum_{t=1}^{N_h} \left(x_{t+h} - \frac{1}{N_h} \sum_{s=1}^{N_h} x_{s+h} \right)^2} \tag{A.3}$$

$$\text{sRMSE}(h) = \frac{\text{RMSE}(h)}{\widehat{\sigma}_y(h)} \tag{A.4}$$

Table A.3 reports the sample standard deviation of the target variable at each forecast horizon. The results reveal that for both indicators there is a decrease in the standard deviation as the forecast horizon increases. This effect is more pronounced for GDP. This pattern suggests that the higher volatility early in the sample may be influencing forecast difficulty.

Table A.3: Target Variable Standard Deviation by Forecast Horizon

Indicator	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 12$
GDP	0.0045	0.0041	0.0041	0.0038	0.0033
Inflation	0.0029	0.0026	0.0027	0.0028	0.0025

Notes: Target variable standard deviation (see Equation A.3). Sample size varies by horizon: 39 observations at h=1, decreasing to 28 at h=12 due to 2019 cutoff. Lower values indicate more stable/predictable economic conditions.

To account for the target volatility, the scaled RMSE results are shown in Table A.4. These show a smaller decrease with horizon compared to the unscaled RMSE, but for Inflation there is still a slight downward trend, especially for the ABM and AR models. One possible explanation is that the early part of the forecasting period represents a structural break (the great recession aftermath and European debt crisis), while the later periods returned to a more predictable regime.

Table A.4: Scaled Root Mean Squared Error (sRMSE)

Model	Indicator	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 12$
ABM	GDP	1.121	1.069	1.061	1.078	1.154
	Inflation	1.038	1.055	1.021	1.022	0.992
DSGE	GDP	1.126	1.110	1.069	1.034	1.147
	Inflation	1.177	1.161	1.135	1.005	1.142
BVAR	GDP	1.065	1.005	1.040	1.083	1.058
	Inflation	1.003	1.072	1.061	1.035	0.998
AR	GDP	1.096	1.104	1.091	1.126	1.246
	Inflation	1.041	1.031	1.045	1.035	1.010
Naive	GDP	1.452	1.445	1.698	1.966	2.015
	Inflation	1.434	1.555	1.695	1.596	1.508

Notes: Scaled RMSE normalizes RMSE by the target variable's standard deviation at each horizon (see Equation A.4). Values closer to 1 indicate forecast errors comparable to target variable volatility. Lower values indicate better accuracy.

B Comparisons of Forecast Accuracy

To directly compare forecast accuracy between models, I apply the Modified Diebold-Mariano (MDM) tests (Diebold and Mariano 1995; Harvey et al. 1997) to each pair of models. The null hypothesis is that two models have equal predictive accuracy under squared-error loss. I use HAC-robust standard errors to account for autocorrelation in loss differentials.

The standard Diebold-Mariano test statistic tends to over-reject the null hypothesis of equal predictive accuracy in finite samples, particularly for multi-step-ahead forecasts (Harvey et al. 1997). This over-rejection occurs because the asymptotic normal approximation performs poorly when the forecast horizon h is large relative to the sample size n.

Harvey et al. (1997) propose a small-sample correction that scales the DM test statistic:

$$S_1^* = S_1 \cdot \sqrt{\frac{n + 1 - 2h + h(h-1)/n}{n}} \tag{B.1}$$

where S_1 is the original DM statistic, n is the number of forecast observations, and h is the forecast horizon. The corrected statistic S_1^* is then compared to a Student-t distribution with $(n-1)$ degrees of freedom rather than the standard normal.

This correction factor is always less than or equal to one (with equality only at $h = 1$ with a large sample), reducing the magnitude of the test statistic and thus producing larger p-values. The resulting adjustments used here are shown in Table B.1.

Table B.1: Harvey–Leybourne–Newbold small-sample correction factors

Horizon (h)	Sample size (n)	Correction factor
1	39	≈ 0.99
2	38	≈ 0.96
4	36	≈ 0.90
8	32	≈ 0.77
12	28	≈ 0.59

The results are presented in Table B.2. While they show some (unadjusted)

significant differences in accuracy between models, these wash out after multiple testing correction is applied (not shown). Due to the small sample size and resulting low power of these tests, these non-rejections should not be interpreted as evidence of equal predictive accuracy.

Table B.2: Pairwise Modified Diebold-Mariano Tests

Model A	Model B	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 12$
Panel A: GDP						
ABM	DSGE	-0.05 (0.964)	-0.67 (0.504)	-0.11 (0.916)	0.74 (0.464)	0.10 (0.922)
ABM	BVAR	0.46 (0.649)	0.66 (0.516)	0.19 (0.850)	-0.02 (0.981)	0.28 (0.784)
ABM	AR	2.26 (0.029**)	-1.75 (0.088*)	-0.85 (0.401)	-0.97 (0.337)	-0.91 (0.371)
ABM	Naive	-1.47 (0.150)	-1.56 (0.126)	-1.61 (0.116)	-1.07 (0.292)	-1.67 (0.107)
DSGE	BVAR	0.33 (0.746)	0.92 (0.363)	0.18 (0.857)	-0.19 (0.848)	0.22 (0.824)
DSGE	AR	0.26 (0.797)	0.10 (0.918)	-0.39 (0.696)	-2.20 (0.036**)	-1.25 (0.223)
DSGE	Naive	-1.71 (0.096*)	-1.49 (0.145)	-1.63 (0.112)	-1.06 (0.296)	-1.64 (0.112)
BVAR	AR	-0.26 (0.798)	-0.93 (0.358)	-0.37 (0.714)	-0.17 (0.867)	-0.43 (0.669)
BVAR	Naive	-1.57 (0.124)	-1.90 (0.066*)	-1.67 (0.104)	-1.22 (0.233)	-2.67 (0.013**)
AR	Naive	-1.53 (0.135)	-1.44 (0.158)	-1.57 (0.125)	-1.00 (0.327)	-1.37 (0.183)
Panel B: Inflation						
ABM	DSGE	-2.18 (0.035**)	-1.38 (0.176)	-1.62 (0.115)	0.22 (0.827)	-1.03 (0.312)
ABM	BVAR	0.38 (0.706)	-0.36 (0.724)	-1.01 (0.319)	-0.62 (0.537)	-0.22 (0.828)
ABM	AR	-0.25 (0.806)	1.40 (0.168)	-1.43 (0.163)	-0.64 (0.527)	-0.76 (0.455)
ABM	Naive	-2.11 (0.042**)	-2.48 (0.018**)	-2.34 (0.025**)	-2.88 (0.007***)	-1.57 (0.129)
DSGE	BVAR	1.52 (0.136)	0.94 (0.352)	0.94 (0.354)	-0.33 (0.746)	1.11 (0.279)
DSGE	AR	2.02 (0.051*)	1.54 (0.132)	1.40 (0.170)	-0.36 (0.724)	1.02 (0.319)
DSGE	Naive	-1.54 (0.131)	-2.02 (0.051*)	-2.22 (0.033**)	-2.60 (0.014**)	-1.26 (0.219)
BVAR	AR	-0.41 (0.686)	1.01 (0.321)	0.37 (0.715)	0.03 (0.974)	-0.98 (0.337)
BVAR	Naive	-1.97 (0.057*)	-2.23 (0.032**)	-2.19 (0.035**)	-2.87 (0.007***)	-1.52 (0.140)
AR	Naive	-2.06 (0.047**)	-2.54 (0.015**)	-2.33 (0.026**)	-2.90 (0.007***)	-1.50 (0.144)

Notes: Modified Diebold-Mariano test statistic for comparing forecast accuracy. Positive values indicate Model B has better accuracy (lower MSE) than Model A. Standard errors computed using Bartlett kernel (Newey-West) HAC estimator with Harvey-Leybourne-Newbold small-sample correction. Unadjusted p-values in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C Stationarity

I verify stationarity of the log-differenced variables, GDP growth and inflation as well as the five market-based indicators used by the BVAR model and the unforecastability tests, using the Augmented Dickey-Fuller (ADF) test (Dickey and Fuller 1979; Said and Dickey 1984). Table C.1 presents the results.

Table C.1: Augmented Dickey-Fuller Stationarity Tests

Indicator	ADF Statistic	p-value	Lags
GDP	-3.393	0.0112**	11
Inflation	-5.196	0.0000***	2
10Y Bond Yield	-7.661	0.0000***	1
ATX	-8.295	0.0000***	0
Oil Price	-9.346	0.0000***	1
EUR/USD	-10.313	0.0000***	0
3M Interbank Rate	-4.731	0.0001***	4

Notes: Augmented Dickey-Fuller (ADF) test for stationarity with constant but no trend. The null hypothesis is that the series contains a unit root (non-stationary). Lag length selected automatically using AIC. GDP, ATX, Oil Price and EUR/USD are log-differenced; Inflation is the log differenced GDP deflator. 10Y Bond Yields and 3M Interbank Rate are first-differenced. All series span 1996Q2–2019Q4. Significance: *** $p < 0.01$, ** $p < 0.05$. Rejection of the null indicates stationarity.

The ADF test evaluates the null hypothesis that the series contains a unit root (non-stationary) against the alternative of stationarity. I use the specification with a constant but no deterministic trend, as log-differenced series are not expected to exhibit trending behaviour. The lag length is selected automatically using the Akaike Information Criterion (AIC) to account for serial correlation in the test regression. Critical values come from MacKinnon (2010).

All series reject the null hypothesis of a unit root. For GDP growth, the ADF statistic of -3.393 yields a p-value of 0.0112, rejecting non-stationarity at the 5% level. For inflation, the evidence is even stronger: the ADF statistic of -5.196 produces a p-value below 0.0001, rejecting non-stationarity at the 1% level. Similar results are obtained for the five market variables which are all able to reject non-stationarity at the 1% level.

These results confirm that the log difference transformations (or in the case of the two interest rates, first difference) have successfully rendered all series stationary.

Table C.2 presents ADF test results for the forecasted series and the forecast errors from each model. For comparison the bottom of the table also shows the ADF results for the actual series constrained to the sample period for each horizon. For the shorter horizons, more samples are available, and non-stationarity can be rejected. However, as the horizon increases and the sample size decreases, the power of the ADF test diminishes. Since the actual series are known to be stationary in a larger sample (from Table C.1), the failure to reject non-stationarity is likely due to low power rather than true non-stationarity.

Table C.2: Augmented Dickey-Fuller Tests: Forecasts and Forecast Errors

Model	h	GDP Growth		Inflation	
		Forecast	Error	Forecast	Error
ABM	1	-2.06 (0.262)	-2.64 (0.085*)	-4.36 (0.000***)	-8.36 (0.000***)
	2	-4.33 (0.000***)	-2.72 (0.071*)	-4.68 (0.000***)	-2.63 (0.087*)
	4	-2.87 (0.049**)	-1.54 (0.515)	-4.06 (0.001***)	-1.11 (0.710)
	8	-0.54 (0.884)	-1.57 (0.498)	-3.72 (0.004***)	-1.90 (0.334)
	12	-1.19 (0.678)	-1.87 (0.344)	0.53 (0.986)	-2.98 (0.037**)
DSGE	1	-4.03 (0.001***)	-2.86 (0.050*)	-6.26 (0.000***)	-2.65 (0.082*)
	2	-4.32 (0.000***)	-3.16 (0.022**)	-3.78 (0.003***)	-1.07 (0.726)
	4	-2.44 (0.132)	-1.88 (0.341)	-2.22 (0.199)	-0.71 (0.844)
	8	-1.09 (0.720)	-1.78 (0.391)	-2.13 (0.231)	-4.27 (0.000***)
	12	-2.83 (0.054*)	-1.61 (0.477)	-1.86 (0.349)	-2.98 (0.037**)
BVAR	1	-4.62 (0.000***)	-3.99 (0.001***)	-8.26 (0.000***)	-5.32 (0.000***)
	2	-4.72 (0.000***)	-3.67 (0.005***)	-3.44 (0.010***)	-2.05 (0.267)
	4	-2.39 (0.144)	-1.63 (0.465)	-3.74 (0.004***)	-1.03 (0.744)
	8	-1.47 (0.550)	-1.57 (0.499)	-1.57 (0.497)	-1.89 (0.335)
	12	-2.81 (0.057*)	-1.75 (0.404)	-0.96 (0.766)	-2.99 (0.036**)
AR	1	-3.25 (0.017**)	-2.65 (0.083*)	-2.77 (0.063*)	-2.08 (0.251)
	2	-5.93 (0.000***)	-2.75 (0.066*)	-4.18 (0.001***)	-2.67 (0.079*)
	4	-2.16 (0.219)	-1.56 (0.504)	-5.41 (0.000***)	-2.77 (0.063*)
	8	-3.39 (0.011**)	-1.58 (0.492)	-3.70 (0.004***)	-1.89 (0.338)
	12	-1.20 (0.672)	-1.86 (0.351)	-0.88 (0.795)	-2.91 (0.044**)
Naive	1	-2.56 (0.102)	-9.93 (0.000***)	-6.33 (0.000***)	-4.09 (0.001***)
	2	-1.03 (0.741)	-3.61 (0.006***)	-6.22 (0.000***)	-2.38 (0.148)
	4	-0.58 (0.875)	0.14 (0.969)	-6.04 (0.000***)	-3.85 (0.002***)
	8	0.55 (0.986)	-5.74 (0.000***)	-5.63 (0.000***)	-2.87 (0.049**)
	12	-1.82 (0.373)	-1.46 (0.554)	-3.53 (0.007***)	-6.57 (0.000***)
		Actuals		Actuals	
	1	-2.87 (0.049**)		-7.53 (0.000***)	
	2	-2.88 (0.047**)		-2.70 (0.074*)	
	4	-1.77 (0.396)		-2.71 (0.072*)	
	8	-1.60 (0.483)		-2.15 (0.223)	
	12	-1.65 (0.456)		-2.81 (0.057*)	

Notes: Augmented Dickey-Fuller (ADF) test for stationarity of forecasts and forecast errors, by model-horizon-indicator. Tests use constant but no trend specification. Lag length selected using AIC. The null hypothesis is unit root (non-stationary). Rejection suggests forecasts/forecast errors are stationary. The table also reports ADF tests on actual values at horizon-specific date ranges (h1: 2010Q2–2019Q4, h2: 2010Q3–2019Q4, h4: 2011Q1–2019Q4, h8: 2012Q1–2019Q4, h12: 2013Q1–2019Q4), matching the evaluation period for each forecast horizon. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D Multiple Testing Corrections

When conducting many simultaneous hypothesis tests, the probability of spurious significance increases mechanically. For example, with $m = 100$ true null hypotheses tested at level $\alpha = 0.05$, the expected number of false rejections is $m\alpha = 5$.

To address this multiplicity, I apply the Holm–Bonferroni step-down procedure (Holm 1979) and report adjusted $\tilde{p}$-values throughout. I treat multiple-testing adjustment as part of the inferential mapping from test statistics to textual claims. Accordingly, I define a family of hypotheses as the set of tests over which I will make a single combined or comparative statement (e.g., "bias is pervasive across indicators and horizons" or "unbiasedness is generally rejected for model A, but not for model B"). In practice, because I frequently compare results across models, indicators, and horizons within each test class, I take each test class (e.g., unbiasedness, efficiency, encompassing) across all corresponding model–indicator–horizon combinations to form one family.

Within a family of m tests, let $P_{(1)} \leq P_{(2)} \leq \cdots \leq P_{(m)}$ denote the ordered (unadjusted) p-values. The Holm-adjusted $\tilde{p}$-values are computed as

$$\tilde{p}_{(k)} = \min\{1,\ \max(\tilde{p}_{(k-1)},\ (m-k+1)\,P_{(k)})\}\,, \qquad \tilde{p}_{(0)} = 0. \tag{D.1}$$

A hypothesis is rejected at level α if $\tilde{p}_{(k)} \leq \alpha$. This procedure controls the family-wise error rate (FWER), i.e. the probability of at least one false rejection within the family, at the chosen level α.

E Additional Notes on Forecast Bias Testing

For multi-step-ahead forecasts ($h > 1$), forecast errors are serially correlated because overlapping horizons reuse the same information, inducing autocorrelation up to lag $h - 1$ (Newey and West 1987). Using standard OLS inference therefore understates uncertainty and over-rejects the null. To obtain valid inference, I use Newey-West HAC-robust covariance estimation, with a minimum bandwidth of $h-1$ to capture the mechanically induced autocorrelation (Newey and West 1987; West 1996).

Poledna et al. (2023) compute p-values using OLS under iid errors[7], which is inappropriate for overlapping forecasts and leads to downward-biased p-values. At horizon $h = 12$, for example, the OLS procedure treats errors 12 quarters apart as independent despite sharing 11 quarters of overlapping information. In addition, I apply a Holm-Bonferroni correction to control for multiple testing (see Appendix D). My $\tilde{p}$-values are therefore more conservative than those reported in (Poledna et al. 2022). This is why my approach fails to reject unbiasedness in most cases.

[7] See `mztest.m` in their replication package.

F Forecast Encompassing Robustness Check

The encompassing regressions in Section 5.3 exclude an intercept term, restricting admissible forecast combinations to the linear span of the individual forecasts. This follows the approach of Diebold (2006). As a robustness check, I also estimate the regressions with an intercept, following the strong encompassing test framework of Timmermann (2006).

The with-intercept specification allows for ex post bias correction:

$$x_t = \alpha + \beta_a \hat{x}^a_{t|t-h} + \beta_b \hat{x}^b_{t|t-h} + u_{t|t-h}, \qquad H_0 : \ \beta_a = 0 \tag{F.1}$$

$$x_t = \alpha + \sum_{i=1}^{K} \beta_i \hat{x}^i_{t|t-h} + u_{t|t-h}, \qquad H_0 : \ \beta_i = 0 \tag{F.2}$$

The results from these regressions are shown in Tables F.1 and F.2. In both the pairwise and joint with-intercept specifications, none of the ABM coefficients are statistically distinguishable from zero after multiple testing adjustment, suggesting that some of the variation is absorbed by the intercept, further reducing the power to identify the model coefficients. As a result, the with-intercept specification cannot clearly discriminate between redundancy and informativeness for the ABM. This is weaker than the no-intercept results, where the ABM showed coefficients significantly different from zero in 8 out of 25 tests for inflation.

Table F.1: Pairwise Encompassing Tests (With Intercept): ABM vs Other Models

Other Model	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 12$
Panel A: GDP					
DSGE	0.35 (1.000)	0.72 (1.000)	-1.27 (1.000)	-1.97 (1.000)	-1.74 (1.000)
BVAR	-1.61 (1.000)	-0.47 (1.000)	-1.39 (1.000)	0.67 (1.000)	-0.63 (1.000)
AR	-4.44 (1.000)	1.21 (1.000)	-1.50 (1.000)	-0.60 (1.000)	-1.44 (1.000)
Naive	-1.64 (1.000)	-0.56 (1.000)	-2.14 (1.000)	-1.85 (1.000)	-2.36 (1.000)
Panel B: Inflation					
DSGE	-1.20 (1.000)	-1.82 (1.000)	0.86 (1.000)	1.99 (1.000)	-0.10 (1.000)
BVAR	-0.79 (1.000)	-1.98 (1.000)	0.79 (1.000)	2.73 (1.000)	2.67 (1.000)
AR	-0.63 (1.000)	-2.64 (1.000)	2.05 (1.000)	1.70 (1.000)	3.53 (1.000)
Naive	-3.76 (1.000)	-1.90 (1.000)	2.06 (1.000)	1.04 (1.000)	0.96 (1.000)

Notes: Pairwise forecast encompassing test, see equation F.1. The table reports the estimated coefficient on the ABM forecast from a regression including the ABM and the indicated benchmark model. The null hypothesis is $\beta_a = 0$, implying that the ABM provides no incremental predictive information conditional on the benchmark forecast. Holm–Bonferroni adjusted $\tilde{p}$-values in parentheses (all values in this table are 1.000 and therefore not statistically significant).

Table F.2: Joint Encompassing Tests (With Intercept): ABM Coefficients

Indicator	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 12$
GDP	-0.90 (1.000)	1.11 (1.000)	-1.18 (1.000)	0.55 (1.000)	-0.78 (1.000)
Inflation	-2.56 (1.000)	-2.94 (0.891)	2.23 (1.000)	3.50 (0.264)	2.43 (1.000)

Notes: ABM coefficients from joint forecast encompassing test, see equation F.2. The table reports estimated coefficient β_i, where model i is the ABM, across forecast horizons h for GDP and inflation. The null hypothesis is $\beta_i = 0$, implying that the ABM provides no incremental predictive information conditional on all other forecasts. Holm–Bonferroni adjusted $\tilde{p}$-values in parentheses. Significance: *** $\tilde{p} < 0.01$, ** $\tilde{p} < 0.05$, * $\tilde{p} < 0.10$.

G Encompassing Test Tables With All Models

This appendix presents tables of results for all model combinations from the encompassing tests defined in Section 5.3 and Appendix F. Table G.1 and Table G.2 present the joint encompassing test results for all models with no-intercept and with-intercept specifications, respectively. Significance values in these tables use unadjusted p-values. Table G.3 and Table G.4 present the pairwise no-intercept results for GDP and inflation, respectively. Table G.5 and Table G.6 present the pairwise with-intercept results for GDP and inflation, respectively.

Table G.1: Joint Encompassing Tests (No Intercept)

Model	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 12$
Panel A: GDP					
ABM	-1.50 (0.570)	2.06 (0.304)	-1.10 (0.467)	-1.10 (0.257)	-0.92 (0.697)
DSGE	0.81 (0.055*)	0.68 (0.125)	0.94 (0.262)	2.19 (0.072*)	3.72 (0.057*)
BVAR	0.22 (0.595)	0.56 (0.431)	1.23 (0.289)	1.40 (0.374)	0.40 (0.808)
AR	3.01 (0.219)	-2.11 (0.204)	-0.36 (0.909)	-2.19 (0.463)	-1.27 (0.011**)
Naive	-0.27 (0.206)	-0.01 (0.959)	-0.08 (0.717)	-0.15 (0.577)	-0.10 (0.456)
R^2	0.189	0.111	0.077	0.280	0.232
n	39	38	36	32	28
Panel B: Inflation					
ABM	0.24 (0.925)	-1.64 (0.329)	3.70 (0.027**)	4.54 (0.002***)	3.10 (0.138)
DSGE	0.22 (0.618)	0.10 (0.699)	-0.20 (0.470)	0.54 (0.183)	-0.43 (0.290)
BVAR	0.60 (0.239)	-0.81 (0.291)	-0.78 (0.541)	-3.21 (0.368)	3.40 (0.360)
AR	0.21 (0.940)	3.69 (0.025**)	-1.16 (0.584)	-0.52 (0.895)	-4.92 (0.023**)
Naive	-0.23 (0.488)	-0.10 (0.388)	-0.31 (0.122)	-0.10 (0.466)	0.13 (0.224)
R^2	0.026	0.061	0.169	0.112	0.150
n	39	38	36	32	28

Notes: Joint forecast encompassing test, see equation 11. The table reports estimated coefficients β_i for each model i across forecast horizons h. The null hypothesis is $\beta_i = 0$, implying that model i provides no incremental predictive information conditional on all other forecasts. Unadjusted p-values in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table G.2: Joint Encompassing Tests (With Intercept)

Model	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 12$
Panel A: GDP					
ABM	-0.90 (0.792)	1.11 (0.701)	-1.18 (0.390)	0.55 (0.737)	-0.78 (0.772)
DSGE	0.88 (0.068*)	0.39 (0.536)	0.50 (0.608)	0.22 (0.806)	3.37 (0.098*)
BVAR	0.24 (0.593)	0.47 (0.478)	-0.94 (0.649)	-4.56 (0.232)	-0.34 (0.942)
AR	3.16 (0.215)	-3.00 (0.166)	-0.10 (0.972)	-1.52 (0.559)	-1.08 (0.316)
Naive	-0.29 (0.240)	0.04 (0.782)	-0.10 (0.658)	-0.16 (0.512)	-0.08 (0.494)
R^2	0.192	0.127	0.105	0.372	0.233
n	39	38	36	32	28
Panel B: Inflation					
ABM	-2.56 (0.389)	-2.94 (0.099*)	2.23 (0.395)	3.50 (0.026**)	2.43 (0.198)
DSGE	0.05 (0.905)	-0.05 (0.846)	-0.35 (0.197)	0.08 (0.915)	-0.79 (0.056*)
BVAR	0.47 (0.377)	-1.26 (0.121)	-1.20 (0.252)	-5.45 (0.020**)	-0.32 (0.937)
AR	-2.78 (0.285)	1.90 (0.298)	-2.20 (0.354)	-0.95 (0.793)	-4.49 (0.036**)
Naive	0.84 (0.062*)	0.01 (0.919)	-0.28 (0.182)	-0.08 (0.461)	0.19 (0.174)
R^2	0.135	0.128	0.202	0.132	0.205
n	39	38	36	32	28

Notes: Joint forecast encompassing test, see equation F.2. The table reports estimated coefficients β_i for each model i across forecast horizons h. The null hypothesis is $\beta_i = 0$, implying that model i provides no incremental predictive information conditional on all other forecasts. Unadjusted p-values in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table G.3: Pairwise Encompassing Tests (No Intercept): GDP

Model A	Model B	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 12$
ABM		1.69 (0.008***)	1.18 (0.016**)	0.47 (0.427)	-1.20 (0.082*)	-1.38 (0.012**)
	DSGE	0.69 (0.027**)	0.83 (0.050*)	1.50 (0.058*)	3.46 (0.000***)	4.01 (0.000***)
ABM		1.30 (0.193)	0.51 (0.759)	-2.08 (0.341)	-1.74 (0.235)	-2.75 (0.104)
	BVAR	0.37 (0.273)	0.59 (0.419)	1.83 (0.094*)	1.58 (0.041**)	2.11 (0.009***)
ABM		-2.67 (0.294)	3.19 (0.046**)	0.75 (0.701)	1.74 (0.122)	2.54 (0.060*)
	AR	4.60 (0.061*)	-1.64 (0.303)	0.78 (0.734)	-0.63 (0.696)	-1.44 (0.222)
ABM		2.07 (0.007***)	1.52 (0.002***)	1.46 (0.019**)	1.63 (0.002***)	1.52 (0.053*)
	Naive	0.04 (0.840)	0.10 (0.480)	-0.07 (0.709)	-0.23 (0.368)	-0.01 (0.971)
DSGE		0.58 (0.112)	0.74 (0.071*)	1.30 (0.133)	3.25 (0.001***)	4.78 (0.001***)
	BVAR	0.59 (0.023**)	0.59 (0.003***)	0.33 (0.335)	-0.52 (0.165)	-0.94 (0.008***)
DSGE		0.62 (0.060*)	0.96 (0.018**)	1.51 (0.067*)	3.24 (0.001***)	3.86 (0.000***)
	AR	1.69 (0.007***)	1.10 (0.019**)	0.58 (0.431)	-1.38 (0.109)	-1.76 (0.003***)
DSGE		1.02 (0.002***)	1.38 (0.000***)	1.94 (0.002***)	2.28 (0.000***)	2.47 (0.003***)
	Naive	0.08 (0.696)	0.16 (0.296)	0.04 (0.804)	-0.17 (0.449)	-0.09 (0.560)
BVAR		0.27 (0.363)	1.00 (0.094*)	1.57 (0.147)	2.32 (0.026**)	1.68 (0.000***)
	AR	1.52 (0.072*)	-0.43 (0.759)	-1.98 (0.448)	-4.22 (0.121)	-2.59 (0.015**)
BVAR		0.76 (0.011**)	0.74 (0.000***)	0.87 (0.005***)	0.92 (0.001***)	0.85 (0.013**)
	Naive	0.02 (0.944)	0.09 (0.559)	-0.13 (0.485)	-0.27 (0.280)	-0.07 (0.655)
AR		2.09 (0.003***)	1.46 (0.004***)	1.97 (0.014**)	2.07 (0.001***)	1.93 (0.038**)
	Naive	0.01 (0.948)	0.15 (0.301)	-0.12 (0.538)	-0.22 (0.410)	0.00 (0.979)

Notes: Pairwise forecast encompassing test, see equation 10. The table reports estimated coefficients β_a and β_b for each pairwise comparison. The null hypothesis is $\beta_a/\beta_b = 0$, implying that model a/b provides no additional information beyond the other one. Unadjusted p-values in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table G.4: Pairwise Encompassing Tests (No Intercept): Inflation

Model A	Model B	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 12$
ABM		1.28 (0.007***)	1.16 (0.001***)	1.24 (0.001***)	0.60 (0.034**)	1.60 (0.008***)
	DSGE	-0.17 (0.613)	0.02 (0.921)	-0.04 (0.880)	0.59 (0.065*)	-0.38 (0.356)
ABM		0.41 (0.367)	1.35 (0.115)	2.25 (0.035**)	4.61 (0.029**)	2.71 (0.325)
	BVAR	0.63 (0.126)	-0.15 (0.837)	-1.04 (0.303)	-3.37 (0.097*)	-1.55 (0.562)
ABM		0.55 (0.757)	-1.97 (0.248)	3.40 (0.011**)	2.94 (0.073*)	4.25 (0.010***)
	AR	0.54 (0.770)	3.15 (0.066*)	-2.18 (0.097*)	-1.72 (0.298)	-3.04 (0.049**)
ABM		1.36 (0.000***)	1.32 (0.000***)	1.53 (0.000***)	1.34 (0.000***)	0.98 (0.000***)
	Naive	-0.24 (0.334)	-0.11 (0.331)	-0.29 (0.120)	-0.12 (0.143)	0.12 (0.275)
DSGE		0.19 (0.360)	0.18 (0.526)	0.19 (0.502)	0.75 (0.068*)	-0.46 (0.233)
	BVAR	0.78 (0.004***)	0.89 (0.010**)	0.97 (0.005***)	0.43 (0.254)	1.67 (0.003***)
DSGE		-0.09 (0.761)	-0.04 (0.860)	0.10 (0.705)	0.71 (0.017**)	-0.26 (0.326)
	AR	1.21 (0.007***)	1.25 (0.001***)	1.08 (0.004***)	0.47 (0.089*)	1.40 (0.000***)
DSGE		1.03 (0.001***)	0.87 (0.000***)	1.04 (0.000***)	1.14 (0.000***)	0.63 (0.000***)
	Naive	-0.23 (0.459)	0.06 (0.631)	0.06 (0.664)	0.01 (0.941)	0.22 (0.074*)
BVAR		0.64 (0.143)	-0.75 (0.280)	-0.25 (0.869)	0.20 (0.938)	4.78 (0.039**)
	AR	0.40 (0.409)	1.99 (0.015**)	1.43 (0.371)	0.99 (0.707)	-3.66 (0.103)
BVAR		0.96 (0.000***)	1.07 (0.000***)	1.37 (0.000***)	1.29 (0.000***)	0.95 (0.000***)
	Naive	0.01 (0.938)	0.01 (0.954)	-0.20 (0.273)	-0.10 (0.247)	0.12 (0.342)
AR		1.40 (0.000***)	1.32 (0.000***)	1.48 (0.000***)	1.27 (0.000***)	0.93 (0.000***)
	Naive	-0.25 (0.321)	-0.11 (0.379)	-0.27 (0.175)	-0.07 (0.295)	0.13 (0.302)

Notes: Pairwise forecast encompassing test, see equation 10. The table reports estimated coefficients β_a and β_b for each pairwise comparison. The null hypothesis is $\beta_a/\beta_b = 0$, implying that model a/b provides no additional information beyond the other one. Unadjusted p-values in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table G.5: Pairwise Encompassing Tests (With Intercept): GDP

Model A	Model B	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 12$
ABM		0.35 (0.884)	0.72 (0.776)	-1.27 (0.417)	-1.97 (0.051*)	-1.74 (0.250)
	DSGE	0.58 (0.095*)	0.76 (0.131)	0.82 (0.369)	2.73 (0.001***)	3.12 (0.126)
ABM		-1.61 (0.521)	-0.47 (0.862)	-1.39 (0.538)	0.67 (0.701)	-0.63 (0.839)
	BVAR	0.21 (0.575)	0.46 (0.525)	-1.55 (0.517)	-6.96 (0.001***)	-3.34 (0.458)
ABM		-4.44 (0.139)	1.21 (0.602)	-1.50 (0.427)	-0.60 (0.658)	-1.44 (0.359)
	AR	3.59 (0.175)	-3.39 (0.071*)	-1.06 (0.682)	-3.62 (0.087*)	-1.11 (0.177)
ABM		-1.64 (0.516)	-0.56 (0.830)	-2.14 (0.157)	-1.85 (0.047**)	-2.36 (0.043**)
	Naive	0.01 (0.960)	0.12 (0.447)	-0.14 (0.478)	-0.25 (0.255)	-0.09 (0.568)
DSGE		0.54 (0.157)	0.69 (0.120)	0.89 (0.334)	1.16 (0.106)	2.58 (0.051*)
	BVAR	0.08 (0.834)	0.41 (0.471)	-1.73 (0.215)	-5.33 (0.000***)	-3.27 (0.030**)
DSGE		0.62 (0.039**)	0.44 (0.426)	0.93 (0.281)	1.81 (0.020**)	3.66 (0.012**)
	AR	1.84 (0.321)	-2.16 (0.355)	-1.30 (0.525)	-3.21 (0.095*)	-1.82 (0.065*)
DSGE		0.80 (0.025**)	0.64 (0.141)	1.04 (0.324)	2.06 (0.084*)	4.50 (0.007***)
	Naive	-0.22 (0.324)	0.05 (0.735)	-0.07 (0.750)	-0.18 (0.515)	-0.09 (0.549)
BVAR		0.24 (0.519)	0.57 (0.316)	-1.87 (0.305)	-4.99 (0.055*)	-3.56 (0.256)
	AR	1.14 (0.600)	-3.18 (0.101)	-1.25 (0.613)	-1.35 (0.566)	-0.44 (0.813)
BVAR		0.23 (0.565)	0.36 (0.599)	-2.48 (0.081*)	-5.43 (0.001***)	-4.11 (0.005***)
	Naive	-0.03 (0.910)	0.09 (0.556)	-0.11 (0.567)	-0.17 (0.424)	0.00 (0.994)
AR		0.93 (0.672)	-2.94 (0.124)	-1.83 (0.434)	-3.58 (0.022**)	-2.09 (0.023**)
	Naive	0.00 (0.991)	0.12 (0.379)	-0.09 (0.660)	-0.22 (0.262)	-0.06 (0.754)

Notes: Pairwise forecast encompassing test, see equation F.1. The table reports estimated coefficients β_a and β_b for each pairwise comparison. The null hypothesis is $\beta_a/\beta_b = 0$, implying that model a/b provides no additional information beyond the other one. Unadjusted p-values in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table G.6: Pairwise Encompassing Tests (With Intercept): Inflation

Model A	Model B	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 12$
ABM		-1.20 (0.483)	-1.82 (0.104)	0.86 (0.728)	1.99 (0.442)	-0.10 (0.960)
	DSGE	0.22 (0.582)	-0.10 (0.652)	-0.07 (0.828)	0.77 (0.178)	-0.69 (0.127)
ABM		-0.79 (0.449)	-1.98 (0.110)	0.79 (0.762)	2.73 (0.158)	2.67 (0.345)
	BVAR	0.46 (0.354)	-1.22 (0.105)	-1.56 (0.055*)	-6.51 (0.000***)	-3.20 (0.157)
ABM		-0.63 (0.786)	-2.64 (0.127)	2.05 (0.360)	1.70 (0.311)	3.53 (0.093*)
	AR	-0.05 (0.978)	1.73 (0.392)	-3.33 (0.034**)	-2.19 (0.220)	-3.75 (0.011**)
ABM		-3.76 (0.085*)	-1.90 (0.101)	2.06 (0.361)	1.04 (0.676)	0.96 (0.486)
	Naive	0.61 (0.048**)	0.01 (0.894)	-0.29 (0.115)	-0.12 (0.253)	0.12 (0.281)
DSGE		-0.04 (0.883)	-0.08 (0.732)	-0.13 (0.638)	-0.02 (0.965)	-0.77 (0.082*)
	BVAR	0.43 (0.376)	-1.14 (0.133)	-1.60 (0.054*)	-5.30 (0.035**)	-1.81 (0.141)
DSGE		0.11 (0.711)	-0.11 (0.665)	-0.26 (0.325)	0.48 (0.590)	-0.92 (0.052*)
	AR	-0.83 (0.485)	-0.07 (0.963)	-3.03 (0.132)	-0.68 (0.838)	-2.73 (0.035**)
DSGE		-0.01 (0.973)	-0.07 (0.775)	-0.32 (0.261)	0.54 (0.254)	-0.68 (0.136)
	Naive	-0.03 (0.914)	-0.07 (0.551)	-0.29 (0.148)	-0.07 (0.444)	0.13 (0.287)
BVAR		0.49 (0.323)	-1.15 (0.127)	-1.25 (0.266)	-5.26 (0.015**)	3.31 (0.079*)
	AR	-0.83 (0.373)	0.06 (0.966)	-2.23 (0.335)	0.05 (0.981)	-3.42 (0.129)
BVAR		0.45 (0.352)	-1.16 (0.131)	-1.54 (0.032**)	-5.10 (0.002***)	-1.78 (0.314)
	Naive	-0.07 (0.720)	-0.08 (0.406)	-0.25 (0.209)	-0.02 (0.732)	0.16 (0.286)
AR		-3.29 (0.072*)	0.34 (0.836)	-1.89 (0.361)	-1.69 (0.281)	-2.13 (0.197)
	Naive	0.54 (0.138)	-0.09 (0.507)	-0.20 (0.272)	-0.10 (0.225)	0.16 (0.191)

Notes: Pairwise forecast encompassing test, see equation F.1. The table reports estimated coefficients β_a and β_b for each pairwise comparison. The null hypothesis is $\beta_a/\beta_b = 0$, implying that model a/b provides no additional information beyond the other one. Unadjusted p-values in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

H Additional Notes on ABM and DSGE Forecasting

Macroeconomic forecasting has evolved substantially over recent decades. DSGE models have been a dominant paradigm since the 1990s, often preferred by economists due to their microfoundations and theoretical coherence (Clarida et al. 1999; Smets and Wouters 2007; Christoffel et al. 2011). Aside from this theoretical advantage, however, their forecasting performance does not exceed reduced-form methods such as BVARs (Crump et al. 2025).

ABMs offer an alternative micro-founded approach that relaxes representative-agent assumptions and allows for heterogeneous, interacting agents and decentralized markets (Dawid and Delli Gatti 2018). This micro-level heterogeneity and nonlinearity enables ABMs to generate endogenous crises, distributional dynamics, and other complex phenomena that are difficult to capture in standard DSGE frameworks (Poledna et al. 2023). The price of this realism is typically greater computational cost and reduced analytical tractability. Farmer (2025) provides an overview of the recent developments in the "quantitative" ABMs in economics (ABMs that are calibrated using large amounts of real-world data).

ABMs and DSGE models are both the result of a computational revolution in economics. While ABMs are algorithms that are too complex to be easily written as mathematical functions and are impossible to solve analytically, DSGE models are mathematical functions that rarely have closed-form analytical solutions. They are so complex that we can typically only solve them by approximating them with numerical algorithms and high-performance computers.

Prior to Poledna et al. (2023) perhaps the largest ABM was developed by Deissenberg et al. (2008). Their highly detailed model, EURACE, included geographic heterogeneity and was designed to model the entire European Union. Its purpose was geared towards policy analysis and scenario exploration rather than forecasting. Certain design choices (such as the daily time step) make it less suitable for

macroeconomic forecasting.

Prior work in ABM forecasting, include Monti et al. (2023), who develop a method for learning ABMs from data using machine learning techniques. In their synthetic data experiments (similar to Delli Gatti and Grazzini 2020, but for a housing market rather than the macroeconomy) they achieve better forecasting performance than other techniques.

Outside of economics, there have been several advances within the Epidemiological ABM literature, with forecasting demonstrated by Venkatramanan et al. (2018) and a reverse forecasting method, in which policy interventions are determined that would have to be taken to achieve desired outcomes by Wood et al. (2022).

DSGE models are the common workhorses of modern macroeconomic forecasting. They are built on microfoundations and assume representative agents optimizing intertemporally under rational expectations. DSGE models have been criticized for their inability to capture key macroeconomic phenomena, such as financial crises and persistent unemployment (Poledna et al. 2023). Nevertheless, Breuss (2018) showed that some DSGE models did indeed provide a reasonable model of the Great Recession in Austria.

Although DSGE models often do not replicate fat-tailed outcomes while ABMs do (Ascari et al. 2015), recent work has shown that they can be adapted to replicate this empirical regularity as well (Dave and Sorge 2025). One area in which ABMs distinguish themselves compared to DSGE models is in their modelling of economic networks, which is proving useful for the study of financial system stability (Steinbacher et al. 2021).

A natural next step is to use Bayesian inference techniques not just for calibration (as Wiese et al. 2024, have done), but to also determine whether the models can be simplified without loss of forecasting power (as has been done for DSGEs, for example by Smets and Wouters 2007).